From Gender Wage Gaps to Glass Ceilings: Mechanisms of Labor Market Inequality Among Highly Educated White Women

Three extensions of Bach, Chernozukhov and Splinder (2016)

Salma El Aazdoudi, Maria Micaela Linares Gomez & Rafael Mourouvin

MX DEPP - Ecole Polytechnique – ENSAE

March 30, 2026

Bach, Chernozhukov and Splinder (2016)

Context : A Well-Documented and Persistent Gender Wage Gap

Motivation

- In the United States, full-time women earn **84%** of men's weekly earnings (BLS, 2024).
- Gender convergence in human capital has narrowed the gap significantly.
- The average wage gap for comparable men and women is now estimated around 8–18% (**?**).
- **Economic Cost:** Underutilized human capital, market inefficiency and disincentives to workforce entry and advancement.

A Literature Focused on the Average 'Unexplained Gap'

- Dominated by compositional analysis (i.e., Oaxaca-Blinder decomposition).
- **Explained:** Human capital (education, exp.), industry, and occupation.
- **Unexplained (Residual):** Discrimination, unobserved traits, and omitted relevant factors.

A Gap in the Gender Wage Gap Literature

Understanding **heterogeneity** is crucial to revealing the underlying mechanisms of inequality and optimal policy.

Understanding Complexity Calls for Exploring Heterogeneity

The Challenge of Heterogeneity

- Traditional models struggle with **specification search and statistical inference** in high-dimensional settings.

Bach, Chernozukhov and Splinder (2016)

- **Data** : American Community Survey (2016)
- **Scope:** High-dimensional framework from 16 dimensions (human capital, household, geography) for full-time workers (25–65).
- **Method: Double Post-Lasso** for automated model selection and valid inference.

Empirical Specification

$$\ln w_i = \alpha + \sum_{j=1}^{p_1} (\beta_j \cdot x_{i,j}) \cdot \text{gender}_i + z_i' \delta + \varepsilon_i$$

- β_j captures the **wage penalty** associated with characteristic X_j between comparable men and women.

Addressing Heterogeneity at Scale: Double Selection

Overcoming Post-Selection Bias

- Inference on simple Lasso OLS is often invalid due to **omitted confounding factors**.
- **Double Selection (Belloni et al., 2014):** Ensures first-order local insensitivity to nuisance parameters (*Neyman Orthogonality*).
- **Estimation Procedure:**
  1. For each j, regress $\ln w_i$ on z_i and $x_{i,-j} \cdot \text{gender}_i$ via post-lasso.
  2. Estimate an auxiliary regression for each j : regress $x_{i,j} \cdot \text{gender}_i$ on z_i.
  3. Obtain parameters of interest via OLS regressing $\ln w_i$ on the selected variables in step 1 or in any regression in step 2.
- **Theory-based regularization parameter**.

Result

Approximately unbiased and Gaussian estimators allowing for valid statistical statements despite the high-dimensional specification.

Addressing Heterogeneity at Scale : Joint Inference

Joint Inference & Multiple Testing

- **Multiplier Bootstrap:** Used to approximate the joint distribution of parameters and construct **joint confidence bands**.
- **Global Validity:** Bounding the probability that any single confidence interval misses the true parameter value.
- **Multiple Testing Correction:** Uses the **Romano-Wolf stepdown procedure** to control the Family-Wise Error Rate (FWER), preventing false positives across the 16+ dimensions.

OLS vs Double Lasso: Revealing Heterogeneity

Average estimates hide large differences

- OLS suggests an average residual gender wage gap of about:
  - **17%** for less-educated women
  - **14%** for college-educated women
- Double Lasso reveals strong heterogeneity:
  - **More than 90%** of less-educated women face a wage penalty
  - only **40%** of college-educated women do
- Median gap:
  - **29%** (less-educated)
  - **9%** (college-educated)

Key takeaway

The gender wage gap is **highly heterogeneous**: average estimates hide large differences across women.

What Drives the Wage Gap?

Key determinants identified by Double Lasso

- **Marriage**: increases the wage gap by about **9–12 pp**.
- **Motherhood**:
  - children $\leq 18 \rightarrow$ wage gap increases by about **5 pp**
  - among highly educated women, those with a child ≤ 4 show a lower gap and may even exhibit a **small wage premium** relative to comparable women without children
- **Other drivers**: race, occupation, industry and work intensity also matter. For instance, the wage gap varies by about **4–9 pp across racial groups**.

Key takeaway

Family status explains a large part of the gender wage gap, while race and job characteristics further amplify the heterogeneity.

Three extensions of Bach, Chernozukhov and Splinder (2016)

Framing the Extension: From Average Gaps to Structural Mechanisms

Departure from the Original Paper

- Original paper estimates a generalized gender wage gap.
- We shift focus toward the **structural mechanisms** generating a potential *glass ceiling* among highly educated women.

Why Focus on White Non-Hispanic Women with a Bachelor's Degree?

- Largest documented gender wage penalties occur among White non-Hispanic women.
- Among women with a Bachelor's degree, the wage gap is **selective** ($\approx 40\%$) rather than universal ($\approx 90\%$ for HS).
- High human capital + similar labor market attachment allows cleaner identification of structural mechanisms.

General Research Question

> *Within the high-human-capital group where the gap is most pronounced, which combinations of career and life choices generate the wage penalty?*

Three economic extensions

1. **Heterogeneity & Glass Ceiling:** Who among highly educated women faces the wage penalty, and is it concentrated at the top of the wage distribution? (Double Lasso, GRF)
2. **Flexibility Price:** Does the labor market exhibit convex returns to hours worked, where the hourly wage increases disproportionately for those who supply extreme effort? (DML)
3. **Motherhood Premium:** Is the observed "motherhood premium" real, or is it an artifact of selection bias where only the highest-earning mothers remain in the workforce? (Doubly robust reweighting)

Extension 1 : Heterogeneity & Glass Ceiling

Motivation: Who Faces the Wage Penalty?

Starting Point: Evidence of a Glass Ceiling

- Gender wage gaps among highly educated workers are often **largest at the top of the wage distribution** (Bertand, Goldin and Katz, 2010).
- Prior literature suggests that inequality may not affect all women equally (Blau and Kahn, 2017).

Key Question

Among highly educated women, who actually faces the wage penalty?

- Is the gender penalty **broadly distributed** across women?
- Or is it **concentrated among specific profiles** (family structure, occupation, work intensity)?

Why This Matters

- Identifying **which characteristics generate the penalty** helps uncover the mechanisms behind the glass ceiling.
- Average wage gaps may hide **large heterogeneity** in the penalty.

Parametric Estimation : Double Post-Lasso Effects

Variable	Estimate	Std.Error	t.value	p.value
Married	-0.047	0.010	-4.822	0.000
Children	-0.045	0.010	-4.516	0.000
Metrop.	0.035	0.015	2.320	0.020
Deg:Bio/Life Sci	-0.057	0.034	-1.689	0.091
Deg:Comp/Inform Sci	-0.084	0.034	-2.473	0.013
Deg:Crim Just/Fire Prot	-0.065	0.035	-1.882	0.060
Deg:Hist	-0.086	0.032	-2.674	0.007
Deg:Phys Sci	-0.107	0.032	-3.345	0.001
Deg:Publ Aff/Policy/Soc Wo	-0.092	0.046	-1.986	0.047
Occ:Arts/Design/Entert/Sports/Media	-0.063	0.029	-2.189	0.029
Occ:Bus Operat Spec	-0.084	0.027	-3.074	0.002
Occ:Comput/Math	-0.094	0.028	-3.315	0.001
Occ:Financ Spec	-0.126	0.028	-4.513	0.000
Occ:Healthc Pract/Technic	-0.078	0.029	-2.656	0.008
Occ:Legal	-0.064	0.032	-2.029	0.042
Occ:Managmnt, Bus/Science/Art	-0.080	0.025	-3.197	0.001
Occ:Office/Administr Supp	-0.087	0.027	-3.239	0.001
Occ:Prod	-0.077	0.041	-1.893	0.058
Occ:Protect Serv	-0.078	0.040	-1.956	0.050
Occ:Sales	-0.075	0.027	-2.770	0.006
Ind:BUISREPSERV	-0.034	0.021	-1.654	0.098
Ind:FINANCE	-0.065	0.019	-3.389	0.001
Ind:PROFE	-0.030	0.017	-1.750	0.080
50to59 work hours	0.017	0.010	1.770	0.077
+70 work hours	-0.065	0.027	-2.398	0.016
Years of Studies	0.039	0.013	2.985	0.003
Exp	0.025	0.011	2.228	0.026
Age	-0.037	0.010	-3.638	0.000

Only statistically significant coefficients at 10% are reported. Coefficients pertaining to geographical variables are omitted.

- **Sector choice matters** : Largest gender pay gaps are found by occupation and degree choice, with women being more penalized in traditionally masculine sectors.
- **Longer hours penalize women** : Women working more than 70 hours per week earn 6.5% less than comparable men.
- **Marriage and Parenthood** : Being married or having children reduce women's weekly earnings by around 4% relative to comparable men.

Non-Parametric Estimation : Causal Forest

Step 1 : Removing confounding with nuisance models - local centering

- Estimate nuisance models using regression forests and obtain residualized outcome and treatment : $\check{Y} = Y - \hat{m}(x)$ and $\check{D} = D - \hat{e}(x)$.

Step 2 : Fitting a tree that clusters observations and constructs adaptive weights

Fit random forest with :

- **Splitting criterion** : maximize heterogeneity in treatment effects.
- **Honesty** : half of the sample splits the tree and half estimates for the terminal leaf.
  - Avoids overfitting and gives asymptotically normal predictions out of each tree.

Construct adaptive weights : $\alpha_i(x) = \frac{1}{B}\sum_{b=1}^{B} \frac{\mathbb{1}(\{X_i \in L_b(x), i \in \mathcal{S}_b\})}{|\{i : X_i \in L_b(x), i \in \mathcal{S}_b\}|}$

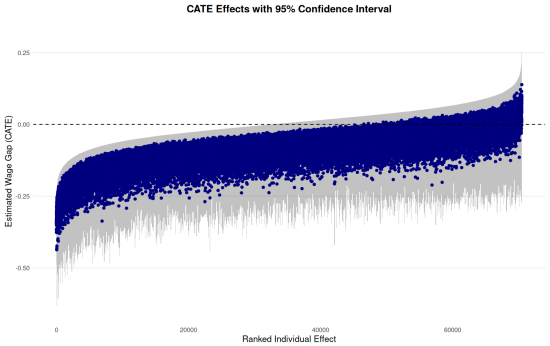
Step 3 : Estimate treatment effects solving local orthogonal moment restriction:

$$\sum_{i=1}^{n} \alpha_i(x)(\check{Y}_i - \tau(x)\check{D}_i)\check{D}_i = 0$$

Step 4 : Compute standard errors using bootstrap of little bags

- Divide trees into small subforests built on different subsamples and estimate prediction variance across these subforests.

Ext.1 Causal Forest : A Highly-Concentrated Wage Gap Penalty



- **Gender wage gap is heterogeneous** across different individuals based on their covariates X.
- For the vast majority of individuals, the estimate remains **consistently below zero**.
- **50% individuals concentrate large and statistically significant gender pay gaps**, which range from 3% to 43.69%.

Ext.1 Causal Forest : Heterogeneity Predictors

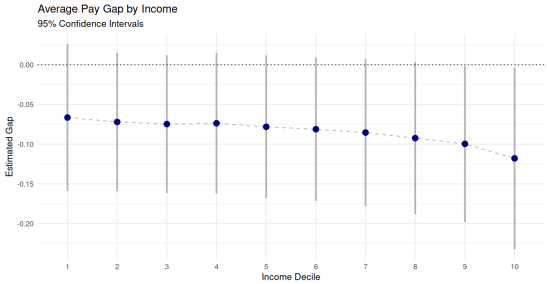
- **Family Structure is Dominant**: Children and being Married are the two most influential factors.
- **Industry Matters**: the gap seems to behave very differently in finance compared to other sectors.
- **Life Stage & Experience**: Gap seems to evolve over a career, but perhaps less dramatically than it does with a change in marital or parental status.

Feature Importance for Heterogeneity

Variable	Importance
Children	0.203
Married	0.153
Ind:Finance	0.149
Age	0.052
Exp2	0.046
Exp	0.040
Occ:Financ Spec	0.022
Deg:Comm	0.018
Deg:Bus	0.015
Region:West South Central Div.	0.014
Ind:Retail	0.013
60-69 work hours	0.012

Features with an importance of the order $10^{-}3$ or smaller are omitted.

Ext.1 Causal Forest : High-Income Women Concentrate Largest Gaps



Note: Dots represent average gender wage gap penalties per income decile. Bars represent average 95% confidence intervals based on average variance estimates per decile

- Gender pay gap **widens as income increases**.
- The relationship is fairly **stable until around the median**, with values around 7.5%, and then drops significantly.
- The **top 10% richest women suffer from an average gender pay gap of 11.79%**, most estimates being significantly different from zero.

Ext.1 Causal Forest : Covariates Effects through Best Linear Projections

- **Extreme effect of long working hours**: women working long hours earn 21% less than comparable men.
- **Mixed effects of sectors**: women are penalized in traditionally masculine sectors while they hold an advantage in diverse areas.
- **Family structure** : married women and women with children earn around 4% less than comparable men.

Variable	Estimate	Std.Error	t.value	p.value
Married	-0.042	0.010	-4.229	0.000
Children	-0.043	0.010	-4.131	0.000
Metrop.	0.042	0.016	2.689	0.007
Deg:Phys.Sci	-0.073	0.044	-1.653	0.098
Deg:Publ.Aff.Policy.Soc.Wo	-0.087	0.051	-1.718	0.086
Occ:Comm.Soc.Serv	0.066	0.029	2.290	0.022
Occ:Educ.Training.Libr	0.042	0.024	1.737	0.082
Occ:Prod	0.080	0.046	1.738	0.082
Ind:Agri	-0.103	0.051	-2.029	0.042
Ind:Finance	-0.056	0.020	-2.848	0.004
Ind:Manuf	0.044	0.020	2.250	0.024
Ind:Wholesale	0.053	0.030	1.770	0.077
+70 work hours	-0.209	0.039	-5.352	0.000

Note: Only significant coefficients at 10% are reported. Coefficients pertaining to geographic variables are omitted.

Extension 2 : Flexibility Price & Convex Returns

Motivation: Why Test the Flexibility Price?

Starting point : What we know from Extension 1

- The wage penalty among highly educated women is concentrated at the top of the wage distribution.
- This suggests a structural mechanism operating in high-paying occupations.

Economic Hypothesis: Convex Returns to Long Hours

- In professional sectors, earnings may increase *more than proportionally* with hours worked.
- Workers supplying extreme hours (60–70+) receive disproportionately high pay.

Implication

- If women are less likely to supply extreme hours, or receive lower returns to them, a wage gap will emerge **at the top**.
- The "glass ceiling" may reflect a **flexibility price** rather than uniform discrimination.

Flexibility Price: Partialling-Out via DML

Objective

For each hour category $h \in \{50\text{–}59,\ 60\text{–}69,\ 70+\}$, estimate the wage premium relative to the 40–49h baseline, controlling for rich worker characteristics.

Category-Specific Partially Linear Model (PLR)

For each h:

$$Y_i = \theta_h D_{ih} + g_{0h}(X_i) + \varepsilon_i$$

- Y_i: log weekly wage
- D_{ih}: indicator for belonging to hour category h
- X_i: high-dimensional controls

Partialling-Out (Residualization)

$$\hat{g}_h(X_i) \approx \mathbb{E}[Y_i \mid X_i], \qquad \hat{m}_h(X_i) \approx \mathbb{E}[D_{ih} \mid X_i]$$

$$\tilde{Y}_i = Y_i - \hat{g}_h(X_i), \qquad \tilde{D}_{ih} = D_{ih} - \hat{m}_h(X_i)$$

Built-in Causal Inference : How PLR allows for Neyman Orthogonality

Orthogonal Score Function

$$\psi(W_i; \theta, \eta) = \big(Y_i - g(X_i)\big)\big(D_i - m(X_i)\big) - \theta\big(D_i - m(X_i)\big)^2$$

where $\eta = (g, m)$ are nuisance functions.

Estimator $\hat{\theta}$ solves:

$$\frac{1}{n}\sum_{i=1}^{n} \psi(W_i; \hat{\theta}, \hat{\eta}) = 0$$

Neyman Orthogonality Property

$$\frac{\partial}{\partial \eta}\mathbb{E}[\psi(W_i; \theta_0, \eta)]\Big|_{\eta=\eta_0} = 0$$

- Small estimation errors in $\hat{g}$ and $\hat{m}$ have only **second-order** effects on $\hat{\theta}$
- Ensures valid inference despite high-dimensional ML estimation
- θ identifies the causal "flexibility price"

Implementation in R: DML Workflow (1/2)

1. **Load required packages**

```r
library(DoubleML)
library(mlr3)
library(mlr3learners)
```

2. **Define the model variables**

```r
covariates = c("married", "child39", "region", "msa", "deg", "occ", "ind", "exp", "exp2", "yos")
y_var = 'lwage'
baseline = '40to49'
greedy_categories = c("50to59", "60to69", "70plus")
```

3. **Specify learners for nuisance functions**
   Random Forest for $\hat{g}(X) \approx \mathbb{E}[Y \mid X]$ and $\hat{m}(X) \approx \mathbb{E}[D \mid X]$

```r
lgbm_reg <- lrn("regr.ranger", num.trees = 500)
lgbm_class <- lrn("classif.ranger", num.trees = 500)
```

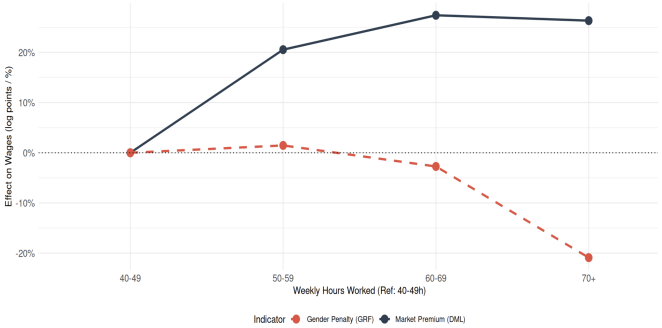
Implementation in R: DML Workflow (2/2)

4. **Declare the `DoubleMLData` object and cross-fitting**

```
dml_plr <- DoubleMLPLR$new(dml_data,
                           ml_l = lgbm_reg,
                           ml_m = lgbm_class,
                           n_folds = 5)
```

5. **Fit the PLR-DML model and extract $\hat{\theta}$**

Results :Extreme Labor Supply Fails to Close the Wage Gap for Women



- **Returns +20.5%** as soon as 50h are reached **+26%**
- **The Glass Ceiling:** A brutal −20.9% penalty emerges when 70h are reached
- **Net Effect:** For women, the "greedy" bonus is almost entirely offset by the gender penalty.

Extension 3 : Motherhood Premium

Extension: Motherhood Premium

Objective

- In the original paper, highly educated women with very young children appear to face a **smaller wage gap**, sometimes even a premium.
- We test whether this pattern persists once mothers and non-mothers are made comparable on observables.
- Define the treatment:

$$D = 1 \quad \text{if the woman has a child in the household}$$

- Let Y denote log weekly wages and $Y(1), Y(0)$ the potential outcomes.

$$\theta^{FT} = E[Y(1) - Y(0) \mid FT = 1]$$

Objective

Estimate the motherhood effect on wages **among full-time women**, after adjusting for observable differences between mothers and non-mothers.

Step 1: Propensity Score Weighting

Rebalancing mothers and non-mothers

- Estimate the propensity score:

$$e(X) = P(D = 1 \mid X)$$

- X: age, education, experience, marital status, occupation, industry, region, hours worked.
- Construct inverse-probability weights:

$$w_i = \frac{D_i}{e(X_i)} + \frac{1 - D_i}{1 - e(X_i)}$$

- This creates a weighted sample where mothers and non-mothers have similar distributions of X.

Key idea

Weighting removes systematic differences in observables before estimating the motherhood effect.

Step 2: DML-IRM Estimation

Doubly robust treatment effect estimation

- We estimate the treatment effect using **Double Machine Learning – IRM**.
- Two nuisance components are estimated:
  - Propensity score: $m(X) = P(D = 1 \mid X)$
  - Outcome regressions:

$$g_1(X) = E[Y \mid D = 1, X], \qquad g_0(X) = E[Y \mid D = 0, X]$$

- The final estimator combines $m(X)$, $g_1(X)$ and $g_0(X)$ through an **orthogonal score**.
- Cross-fitting is used to avoid overfitting bias.

Why DML?

DML is **doubly robust**: the estimator remains consistent if either the propensity score model or the outcome model is correctly specified.

Step 3: Common Support and Trimming

Restricting the comparison to comparable women

- The propensity score distributions of mothers and non-mothers are separated at the extremes.
- To avoid relying on observations with almost no comparable counterpart, we trim the sample:

$$0.02 < e(X) < 0.98, \quad 0.05 < e(X) < 0.95, \quad 0.10 < e(X) < 0.90$$

- Trimming reduces extreme weights and improves effective sample size.
- Example:
  - with 2% trimming, treated ESS is about **4,255**
  - with 5% trimming, treated ESS rises to about **7,129**

Interpretation

Stricter trimming changes the target population: the effect is then estimated for women located in the region of **common support**, where mothers and non-mothers are genuinely comparable.

Results: Estimated Motherhood Effect

DML-IRM estimates

- No trimming:

$$\hat{\theta} = 0.0191 \Rightarrow \mathbf{+1.93\%}$$

- 2% trimming:

$$\hat{\theta} = 0.0184 \Rightarrow \mathbf{+1.85\%}$$

- 5% trimming:

$$\hat{\theta} = 0.0328 \Rightarrow \mathbf{+3.33\%}$$

- 10% trimming:

$$\hat{\theta} = 0.0500 \Rightarrow \mathbf{+5.13\%}$$

Interpretation

Among comparable full-time women, motherhood is associated with a **small wage premium (2–5%)**. This may reflect **selection into full-time employment** or **heterogeneous returns** for highly attached and highly skilled mothers.